

2. Wine Quality Analysis Report

Title

Wine Quality Analysis using Feature Scaling and PCA

Objective

The objective of this case study is to analyze wine quality using machine learning techniques. Feature scaling and Principal Component Analysis (PCA) are applied to improve model performance and reduce feature complexity.

Dataset

The dataset used is the Wine Quality Dataset obtained from UCI/Kaggle. It contains chemical properties of wine such as acidity, sugar, alcohol, and pH levels, along with quality ratings.

Methodology

The dataset was loaded and explored to understand its features. The target variable was identified as wine quality.

Feature scaling was applied using StandardScaler to normalize the data. PCA was then used to reduce the number of features while preserving important information.

The dataset was split into training and testing sets. A machine learning model was trained and evaluated on the transformed dataset.

Result

The model achieved an accuracy of approximately **69–75%**. PCA successfully reduced feature complexity while maintaining reasonable predictive performance.

Conclusion

This case study highlights the importance of feature scaling and dimensionality reduction in machine learning. PCA helps in reducing complexity while maintaining performance, making models more efficient and easier to train.